KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

MID-REVIEW PROJECT REPORT

Youth Smoking and Drug Abuse Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team 16

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1. Introduction

The project, titled "Youth Smoking and Drug Abuse Analysis", focuses on exploring patterns in a structured drug-consumption dataset through data cleaning, statistical analysis, and visualization. The project follows a data analysis workflow in which raw data is inspected and cleaned, followed by exploratory data analysis (EDA) to understand distributions and relationships among demographic, personality, and drug-consumption variables.

For this project, participants aged 18–24 are considered the youth group. This project definition is used to focus the analysis and does not imply that the original dataset labels all participants as youth.

2. Problem Statement

The dataset contains demographic information, personality scores, and reported consumption levels for several substances. Some records contain missing values, while several variables are represented using encoded numerical values. Direct analysis of such data can make interpretation difficult. Therefore, the project aims to inspect, clean, analyze, and visualize the dataset to identify meaningful patterns and relationships.

3. Objectives

- Understand the structure and characteristics of the Drug Consumption dataset.
- Identify missing values, duplicate records, data types, and unique values.
- Handle missing values using suitable imputation techniques.

- Identify potential outliers using the Z-score method.
- Perform univariate and bivariate exploratory data analysis.
- Select suitable visualizations based on variable types.
- Identify patterns that guide the later stages of the project.

4. Scope of the Project

The current mid-review covers work completed through Week 4: data collection, dataset inspection, data cleaning, data-quality checking, and exploratory data analysis. Later stages will extend this work through feature engineering, feature selection, PCA, and interactive dashboard development.

5. Dataset Description

The project uses the Drug Consumption dataset from the UCI Machine Learning Repository. It is a structured/tabular dataset containing 1,885 records and 32 columns.

Attribute	Description
Dataset	Drug Consumption (UCI)
Records	1,885
Columns	32
Data type	Structured / tabular
Main groups	Demographic, personality and drug-consumption variables
Youth definition	Participants aged 18–24

6. Variable Overview

6.1 Demographic Variables

Age, Gender, Education, Country, and Ethnicity describe demographic characteristics. Several are represented by encoded values in the original dataset.

6.2 Personality Variables

Nscore, Escore, Oscore, Ascore, Cscore, Impulsive, and SS represent personality-related scores: Neuroticism, Extraversion, Openness, Agreeableness, Conscientiousness, Impulsivity, and Sensation Seeking.

6.3 Drug Consumption Variables

The dataset includes Alcohol, Cannabis, Coke, Heroin, Ecstasy, LSD, Meth, Mushrooms, Nicotine and other substance variables. Consumption is represented using coded levels such as CL0–CL6.

7. Methodology / Workflow

- Data Collection – obtain and load the UCI Drug Consumption dataset.
- Know Your Data – inspect rows, columns, shape, data types, statistical summaries, missing values, duplicates and unique values.
- Data Cleaning – create a clean copy, handle missing values and inspect potential outliers.

- EDA – perform univariate and bivariate analysis using suitable plots.
- Next Stages – use the findings to guide feature engineering, dimensionality reduction and dashboard development.

8. Week 1 – Data Collection

During Week 1, the team selected the Drug Consumption dataset from the UCI source and established the project direction around understanding substance-consumption patterns and visualizing demographic and personality information. The dataset provides multiple variable types suitable for data-quality analysis and exploratory visualization.

9. Week 2 – Know Your Data

The dataset was loaded using Pandas and systematically inspected. The first and last rows were viewed using head() and tail(). The shape was checked and found to be 1,885 rows and 32 columns. Column names, data types, statistical summaries, missing-value counts, duplicate records, and unique values were also examined.

The dataset contains 12 float columns, 1 integer column, and 19 object columns. Missing values were observed in Age, Gender, Education, and Nscore, with 113 missing values in each. Duplicate-row checking showed no duplicate records.

Several variables use encoded numerical representations. These encodings were retained during initial inspection because the aim at this stage was to understand the original data structure.

10. Week 3 – Data Cleaning and Data Quality

10.1 Missing-Value Handling

A copy of the original dataset was created for cleaning. Median imputation was applied to Age, Education, and Nscore. Mode imputation was applied to Gender. Median was selected because it is less affected by extreme values and preserves the order or numerical scale of these variables. Gender is categorical, so its missing values were replaced using the most frequent category.

After imputation, the dataset contained zero missing values.

10.2 Encoded Variables and Visualization Labels

Country and other coded variables were not unnecessarily converted during cleaning because they did not have missing values requiring imputation, and the Week 4 analysis did not require detailed interpretation of every encoded category. Nscore had missing values and was therefore handled through median imputation.

For visualization, Age, Gender, and Education were mapped to readable labels without modifying their original columns. Median-imputed Education values were separately labelled as “Median Imputed” for visualization and excluded from the Education-versus-Alcohol category plot.

10.3 Outlier Detection

Potential outliers were investigated using the Z-score method with $|Z| > 3$ as the detection criterion. Potential outliers were observed in variables including Age, Ethnicity, Nscore, Escore, Oscore, Ascore, Cscore, and

Impulsive. They were not automatically removed because a potential outlier can represent a genuine observation.

10.4 Data Quality Status

Check	Result
Records	1,885
Columns	32
Missing values after imputation	0
Duplicate rows	0
Potential outliers	Identified and retained for further consideration

11. Week 4 – Exploratory Data Analysis

EDA was performed to understand individual variable distributions and relationships between pairs of variables. The visualization method was selected according to the variable types.

11.1 Univariate Analysis

Univariate analysis examines one variable at a time. Count plots were used for categorical variables such as Gender, Age Group, Education Level, and drug-consumption categories. Histograms were used for numerical personality scores such as Nscore, Escore, Oscore, Ascore, Cscore, Impulsive, and SS.

Drug-consumption variables were examined using count plots to understand the frequency distribution across consumption-level categories.

11.2 Bivariate Analysis

Bivariate analysis examines two variables. Grouped count plots were used for Age Group versus Gender, Gender versus Alcohol, Age Group versus Alcohol, and Education Level versus Alcohol. Boxplots were used for Nscore versus Alcohol and Nscore versus Cannabis. Scatter plots were used for Nscore versus Escore and Nscore versus Oscore. A correlation heatmap was created for the numerical personality variables.

12. Key EDA Findings

- The dataset contains demographic, personality and drug-consumption variables, requiring different visualization techniques.
- Consumption variables are categorical and represented using CL0–CL6 levels, making count plots suitable for frequency analysis.
- Personality variables are numerical scores and can be examined using histograms, boxplots, scatter plots and correlation analysis.
- Demographic-versus-consumption plots allow comparison of reported consumption-level distributions across groups.
- The correlation heatmap gives an overall view of relationships among personality scores.
- The EDA provides a foundation for feature engineering, dimensionality reduction and dashboard development.

13. Dataset Readiness

After Week 3 cleaning, all identified missing values were imputed and no duplicate rows remained. Potential outliers were identified and retained for further consideration rather than removed automatically. The dataset is prepared for the exploratory analysis completed in Week 4 and can proceed to the subsequent project stages.

14. Challenges Faced

- Understanding encoded demographic and consumption values.
- Selecting suitable imputation techniques for different data types.
- Handling missing values without unnecessarily changing the original data.
- Choosing appropriate visualization types for categorical and numerical variables.
- Interpreting potential outliers carefully instead of removing them automatically.

15. Expected Final Results

The final project is expected to extend the current analysis into a complete data-storytelling workflow. Future stages will include feature engineering and appropriate scaling/encoding, feature selection, PCA and explained-variance analysis, followed by an interactive dashboard. The final output will communicate major patterns and insights through an organized visual narrative.

16. Future Work

- Perform feature engineering and appropriate feature scaling/encoding.
- Analyze correlations and consider feature selection.
- Apply PCA and visualize explained variance.
- Develop interactive visualizations and a narrative dashboard.
- Integrate the completed workflow into the final report and presentation.

17. Conclusion

The first four weeks established a foundation for further analysis. The dataset was inspected, data-quality issues were identified, missing values were handled using suitable imputation techniques, and potential outliers were investigated. Week 4 EDA provided visual understanding of demographic distributions, personality-score distributions, drug-consumption levels, and relationships between selected variables. These results will guide the remaining stages of the project and development of the final data-storytelling dashboard.

18. References

- UCI Machine Learning Repository – https://archive.ics.uci.edu/dataset/373/drug+consumption+quantified.
- Pandas documentation – https://pandas.pydata.org/docs/
- Matplotlib documentation – https://matplotlib.org/stable/
- Seaborn documentation – https://seaborn.pydata.org/
- Scikit-learn documentation – https://scikit-learn.org/stable/